
Time-Dependent System Reliability

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Definition 0.1.1 (Stochastic Process). A stochastic process $\mathbf{X} = \{X(t), t \in \mathcal{T}\}$ is a collection of random variables. That is, for each t in the index set $\mathcal{T}$, $X(t)$ is a random variable.

The index is often interpreted as *time* and, as a result, we refer to $X(t)$ as the *state* of the process at time t . The *state space* of a stochastic process is defined as the set of all possible values that the random variables $X(t)$ can assume. For instance,

- $\{X(t), t = 0, 1, \dots\}$ is a discrete-time stochastic process indexed by the nonnegative integers;
- $\{X(t), t \geq 0\}$ is a continuous-time stochastic process indexed by the nonnegative real numbers.

Thus, a stochastic process is a family of random variables that describes the system evolution through time of some (physical) process. This gives the conceptual fundation for *Time-Dependent System Reliability*.

Markov Property states that given present state, forecast of future is independent of past, i.e.,

$$P\{\text{Futhure} \mid \text{Present, Past}\} = P\{\text{Futhure} \mid \text{Present}\}.$$

Then, a stochastic process that follows the Markov property is called a *Markov process*.

Definition 0.1.2 (Discrete-Parameter Markov Chain). Let $\{X_n, n = 0, 1, \dots\}$ be a stochastic process that takes on a finite or countable number of possible values. Suppose that

$$P\{X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0\} = P\{X_{n+1} = j \mid X_n = i\}$$

for all states $i_0, i_1, \dots, i, j$ and all $n \geq 0$, then stochastic process $\{X_n, n = 0, 1, \dots\}$ is known as a discrete-parameter Markov chain (DPMC). In addition, if $P\{X_{n+1} = j \mid X_n = i\} = p_{ij}$ for any $n \geq 0$, it is a homogeneous DPMC.

In homogeneous DPMC, p_{ij} represents the probability that the process will when in state i , next make a transition into state j , which is call *one-step transition probability*. Let $\mathbf{P}$ denote the matrix of p_{ij} , so that the *transition probability matrix* is

$$P = \begin{bmatrix} p_{00} & p_{01} & p_{02} & \cdots \\ p_{10} & p_{11} & p_{12} & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ p_{i0} & p_{i1} & p_{i2} & \cdots \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix},$$

where $0 \leq p_{ij} \leq 1$ and $\sum_j p_{ij} = 1$ for all i .

Define the n -step transition probability $p_{ij}(n)$ as

$$p_{ij}(n) = P\{X_{n+k} = j \mid X_k = i\}, \quad n \geq 0, i, j \geq 0.$$

Theorem 0.1.1 (Chapman-Kolmogorov Equations). The *Chapman-Kolmogorov Equations* are

$$p_{ij}(n+m) = \sum_{k=0}^{\infty} p_{ik}(n)p_{kj}(m) \quad \forall n, m \geq 0, \forall i, j. \quad (1)$$

Proof.

$$\begin{aligned} p_{ij}(n+m) &= P\{X_{n+m} = j | X_0 = i\} && \leftarrow \text{Definition} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j, X_n = k | X_0 = i\} && \leftarrow \text{Law of total probability} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j | X_n = k, X_0 = i\} P\{X_n = k | X_0 = i\} && \leftarrow \text{Conditional probability} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j | X_n = k\} P\{X_n = k | X_0 = i\} && \leftarrow \text{Markov property} \\ &= \sum_{k=0}^{\infty} p_{ik}(n)p_{kj}(m). \end{aligned}$$

□

If we let $\mathbf{P}(n)$ denote the matrix of n -step transition probability $p_{ij}(n+m)$, then

$$\mathbf{P}(n+m) = \mathbf{P}(n) \cdot \mathbf{P}(m), \quad (2)$$

in particular,

$$\mathbf{P}(2) = \mathbf{P}^2 \quad \mathbf{P}(n) = \mathbf{P}^n.$$

Remark 0.1.1. The matrix of n -step transition probabilities is obtained by multiplying the matrix of one-step transition probabilities by itself $n - 1$ times.

Let $p_i(n)$ denote the probability that a Markov chain is in state i at time n , i.e.,

$$p_i(n) = P\{X_n = i\}.$$

In vector notation, let $\mathbf{p}(n) = (p_1(n), p_2(n), \dots, p_i(n), \dots)$.

The state probabilities at any time step n can be obtained from a knowledge of the initial probability distribution and the matrix of transition probabilities. From the theorem of total probability,

$$p_i(n) = \sum_{\text{all } k} p_k(0) P\{X_n = i | X_0 = k\} = \sum_{\text{all } k} p_k(0) p_{ki}(n),$$

writing in matrix form becomes

$$\mathbf{p}(n) = \mathbf{p}(0)\mathbf{P}(n) = \mathbf{p}(0)\mathbf{P}^n,$$

where $\mathbf{p}(0)$ denotes the initial state distribution.

Definition 0.1.3 (Steady-State Distribution). Let $\mathbf{P}$ be the transition probability matrix of a DPMC, and let the vector $\boldsymbol{\pi}$ be a probability distribution, whose element π_j denote the probability of being in state j , i.e.,

$$0 \leq \pi_j \leq 1, \quad \sum_{\text{all } j} \pi_j = 1.$$

Then $\boldsymbol{\pi}$ is said to be steady-state distribution if and only if

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}.$$

Remark 0.1.2 (Steady-State Distribution & Eigenproblem). Recall the eigenvalue problem

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x},$$

where λ is an eigenvalue of $\mathbf{A}$ and $\mathbf{x}$ is the corresponding right eigenvector.

Since a steady-state distribution satisfies

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi},$$

$\boldsymbol{\pi}$ is a left eigenvector of $\mathbf{P}$ associated with the eigenvalue $\lambda = 1$.

Equivalently,

$$\mathbf{P}^\top \boldsymbol{\pi}^\top = \boldsymbol{\pi}^\top.$$

Thus, the stationary distribution can be computed by solving

$$\begin{cases} \boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \\ \boldsymbol{\pi}\mathbf{e} = 1. \end{cases} \quad (3)$$

where $\mathbf{e}$ is the vector of ones.

Define $f_{ij}(n)$ for $i \neq j$ as the probability that, starting from state i , the first passage to state j occurs in exactly n steps

$$f_{ij}(n) = P\{X_n = j, X_k \neq j, k = 1, \dots, n-1 | X_0 = i\}, \quad n = 1, 2, \dots$$

Proposition 0.1.1 (Relationship between $f_{ij}(n)$ and $p_{ij}(n)$). For $i \neq j$,

$$f_{ij}(n) = p_{ij}(n) - \sum_{l=1}^{n-1} f_{ij}(l)p_{ij}(n-l). \quad (4)$$

When $i = j$,

$$f_{ii}(n) = p_{ii}(n) - \sum_{l=1}^{n-1} f_{ii}(l)p_{ii}(n-l). \quad (5)$$

Proof. If the chain is in state j at step n , then the first visit to state j must occur at some step $\ell \leq n$. If the first visit occurs at step ℓ , then the probability of this event is $f_{ij}(\ell)$. After reaching state j , the probability of being in

state j again after the remaining $n - \ell$ steps is $p_{jj}(n - \ell)$. Therefore,

$$\begin{aligned} p_{ij}(1) &= f_{ij}(1), \\ p_{ij}(2) &= f_{ij}(1)p_{jj}(1) + f_{ij}(2), \\ p_{ij}(3) &= f_{ij}(1)p_{jj}(2) + f_{ij}(2)p_{jj}(1) + f_{ij}(3), \\ &\vdots \\ p_{ij}(n) &= f_{ij}(n) + \sum_{\ell=1}^{n-1} f_{ij}(\ell)p_{jj}(n - \ell). \end{aligned}$$

Rearranging gives

$$f_{ij}(n) = p_{ij}(n) - \sum_{\ell=1}^{n-1} f_{ij}(\ell)p_{jj}(n - \ell).$$

The case $i = j$ follows analogously. □

Proposition 0.1.1 essentially gives the following relationship:

Being in state j at step $n = \sum_{\ell=1}^n (\text{first reaching } j \text{ at step } \ell) \times (\text{being in } j \text{ again after the remaining } n - \ell \text{ steps}),$

and it also gives a recursive way to compute $f_{ij}(n)$ based on $p_{ij}(n)$.

Remark 0.1.3 (How to Calculate $p_{ij}(n)$ and $f_{ij}(n)$). Use CK equation in matrix form for $p_{ij}(n)$, and then use the recursive equation (4) for $f_{ij}(n)$.

Let

$$T_j = \min\{n \geq 1 : X_n = j\}$$

be the first return time to j . Then, we can define the mean recurrence time m_{jj} of state j as

$$m_{jj} = \mathbb{E}[\text{Recurrence time}] = \mathbb{E}[T_j] = \sum_{n=1}^{\infty} n f_{jj}(T_j = n) = \sum_{n=1}^{\infty} n f_{jj}(n),$$

which is the average number of steps taken to return to state j for the first time after leaving it. Similarly, for $i \neq j$, we can define the mean first passage time

$$m_{ij} = \sum_{n=1}^{\infty} n f_{ij}(n).$$

The m_{ij} uniquely satisfy

$$m_{ij} = f_{ij}(1) + \sum_{k \neq j} p_{ik}(1 + m_{kj}) = 1 + \sum_{k \neq j} p_{ik}m_{kj},$$

since the process in state i either goes to state j in one step with probability p_{ij} , or goes first to some intermediate state k in one step with probability p_{ik} and then eventually on to j in an additional m_{kj} steps.

Define $q_{ij}(m)$ as the probability that, starting from state i , state j is reached for the first time within m steps

$$q_{ij}(m) = \sum_{n=1}^m f_{ij}(n).$$

Equivalently,

$$q_{ij}(m) = P\{T_j \leq m \mid X_0 = i\},$$

where T_j denotes the first passage time to state j . Since $q_{ij}(m)$ is the cumulative probability associated with the first passage probabilities $f_{ij}(n)$,

$$f_{ij}(m) = q_{ij}(m) - q_{ij}(m-1).$$

Letting $m \rightarrow \infty$, we define

$$f_{ij} = \sum_{n=1}^{\infty} f_{ij}(n) = \lim_{m \rightarrow \infty} q_{ij}(m),$$

which is the probability that state j is ever reached from state i . When $i = j$,

$$f_{jj} = \sum_{n=1}^{\infty} f_{jj}(n) = \lim_{m \rightarrow \infty} q_{jj}(m),$$

which is the probability that, starting from state j , the process eventually returns to state j .

Proposition 0.1.2 (Relationship among $p_{ij}(n)$, $f_{ij}(n)$, and $q_{ij}(n)$). For $i \neq j$ and $n \geq 1$

$$q_{ij}(n) \geq p_{ij}(n) \geq f_{ij}(n).$$

Proof. We prove this directly from the definitions. Recall that

$$p_{ij}(n) = P(X_n = j \mid X_0 = i),$$

$$f_{ij}(n) = P(X_n = j, X_k \neq j, k = 1, \dots, n-1 \mid X_0 = i),$$

$$q_{ij}(n) = P(\min\{m : X_m = j\} \leq n \mid X_0 = i).$$

First, split $q_{ij}(n)$ according to whether the first visit to j occurs before time n or exactly at time n :

$$q_{ij}(n) = P(\min\{m : X_m = j\} \leq n-1 \mid X_0 = i) + \underbrace{P(\min\{m : X_m = j\} = n \mid X_0 = i)}_{f_{ij}(n)}.$$

Next, split $p_{ij}(n)$ according to whether the chain visits j for the first time at time n , or has already visited j by time $n-1$:

$$p_{ij}(n) = P(X_n = j \mid X_0 = i) = \underbrace{P(\min\{m : X_m = j\} = n \mid X_0 = i)}_{f_{ij}(n)} + P(\min\{m : X_m = j\} \leq n-1, X_n = j \mid X_0 = i).$$

Since

$$\{\min\{m : X_m = j\} \leq n-1, X_n = j\} \subseteq \{\min\{m : X_m = j\} \leq n-1\},$$

we have

$$P(\min\{m : X_m = j\} \leq n-1, X_n = j \mid X_0 = i) \leq P(\min\{m : X_m = j\} \leq n-1 \mid X_0 = i).$$

Therefore, comparing the two decompositions above gives

$$p_{ij}(n) \leq q_{ij}(n).$$

Also, from

$$p_{ij}(n) = f_{ij}(n) + P(\min\{m : X_m = j\} \leq n-1, X_n = j \mid X_0 = i),$$

and the nonnegativity of probability, we obtain

$$p_{ij}(n) \geq f_{ij}(n).$$

Hence,

$$q_{ij}(n) \geq p_{ij}(n) \geq f_{ij}(n).$$

□

Let

$$L_j = \min\{n \geq 1 : X_n \neq j\}$$

be starting from state j , the number of steps until the process first leaves state j . Then, we can define the mean occupation time l_{jj} of state j as

$$l_{jj} = \mathbb{E}[\text{Occupation time}] = \mathbb{E}[L_j] = \sum_{n=1}^{\infty} n p_{jj}^{n-1} (1 - p_{jj}) = \frac{1}{1 - p_{jj}}.$$

Proposition 0.1.3 (Closed-Form Expression of Mean Occupation Time). The closed-form of mean occupation time of state j is

$$l_{jj} = \mathbb{E}[L_j] = \frac{1}{1 - p_{jj}}.$$

Proof. Note that $L_j \sim \text{Geom}(1 - p_{jj})$ and

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}, \quad |x| < 1.$$

Differentiate both sides with respect to x to obtain

$$\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1 - x)^2}.$$

Letting $x = p_{jj}$,

$$\sum_{n=1}^{\infty} n p_{jj}^{n-1} = \frac{1}{(1 - p_{jj})^2}.$$

Therefore,

$$l_{jj} = \sum_{n=1}^{\infty} n p_{jj}^{n-1} (1 - p_{jj}) = (1 - p_{jj}) \sum_{n=1}^{\infty} n p_{jj}^{n-1} = (1 - p_{jj}) \frac{1}{(1 - p_{jj})^2} = \frac{1}{1 - p_{jj}}.$$

□